

AMOGH BIRADAR

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Education

VIT Bhopal University

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Cumulative GPA: 8.01 / 10.0

Bhopal, Madhya Pradesh

Expected May 2026

Technical Skills

Languages: Python, C++ , Java, HTML, CSS, JavaScript, SQL

Technologies: Machine Learning, Data Science, Artificial Intelligence, Natural Language Processing (NLP), OpenCV

Frameworks & Tools: Flask, Tkinter, Git, Microsoft Azure, Power BI, Chart.js

Core Competencies: Data Structures & Algorithms, Cloud Computing, REST APIs, GUI Development, Version Control

Project Experience

Multi-Modal Driver Inattention Detection System

Tech Stack: Python, TensorFlow, YOLOv8, CNN-LSTM, OpenCV

Sep 2025 – Present

- Developed an Integrated Driver Safety System (IDSS) meeting SAE Level 3 standards using sensor fusion and deep learning to detect driver fatigue and distraction.
- Engineered a novel Attention-Based Spatio-Temporal Network (ASTN), achieving 96.54% accuracy and outperforming standard CNN-LSTM baselines by 7% on a 287,000+ frame dataset.
- Integrated YOLOv8n for real-time object detection (safe driving, phone usage) on edge devices.

AI-Powered Health Chatbot

Tech Stack: Python, NLP, Flask, JavaScript, Chart.js, Groq LLM

Jul 2024 – Apr 2025

- Developed an NLP-driven health chatbot with Meta's Llama 3 to provide instant medical guidance to users in remote areas.
- Trained the model on a 5,000+ medical query dataset, achieving 85% intent recognition accuracy.
- Designed a user-friendly UI with Flask and JavaScript, reducing session drop-offs by 30%.
- Implemented dynamic visualizations with Chart.js for tracking user health trends and symptoms.

Sanskrit Text-to-Speech (TTS) System

Tech Stack: Python, Machine Learning, Google Translate API, Microsoft Azure

Feb 2024 – May 2024

- Developed a 3-model ML pipeline for natural language understanding and Sanskrit-to-Roman transliteration.
- Achieved 90% accuracy in Sanskrit TTS synthesis using AI voice models on Microsoft Azure.
- Contributed to a 5-member team by developing the core module for speech synthesis from text.
- Utilized Azure and Google APIs for scalable, cloud-based NLP and multilingual support.

Face Detection & OCR System

Tech Stack: Python, OpenCV, MySQL, Tkinter, Tesseract-OCR

Aug 2023 – Oct 2023

- Built a facial recognition system with 95% accuracy by training an OpenCV model on a 13,000+ image dataset.
- Improved system performance by 3% through algorithm optimization and image pre-processing.
- Designed a GUI with Tkinter and implemented a secure MySQL backend for data management.
- Integrated an OCR module to extract text, improving data entry functionality by 30%.

Achievements & Certifications

- Collaborated with a 4-member team in FOSS Hack 2025 to develop and deploy a prototype of a scalable portfolio optimizer web app, which leverages LLM/NLP for user inputs and integrates a cloud-based GroqAI with Meta Llama 3.3 to achieve faster performance, in a 48-hour timeframe.
- Complete AI, Machine Learning & Data Science Bootcamp (Udemy)
- Oracle Certified Generative AI Professional; Oracle Certified AI Foundations Associate

Additional Information

- Extracurriculars:** Active member of the University Football team, the University Dance Club (Western/Bollywood) and the Ekfrasies Drama Society(Stage Productions).
- Languages:** English (Fluent), Hindi (Fluent), Kannada (Native), Marathi (Conversational), French (Elementary).